

# MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)



**MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)**

**SEMESTER 1 2025/2026**

**PROGRESS REPORT 3 MINI PROJECT**

**SECTION 2**

**GROUP 17**

**LECTURER: ZULKIFLI BIN ZAINAL ABIDIN & WAHJU SEDIONO**

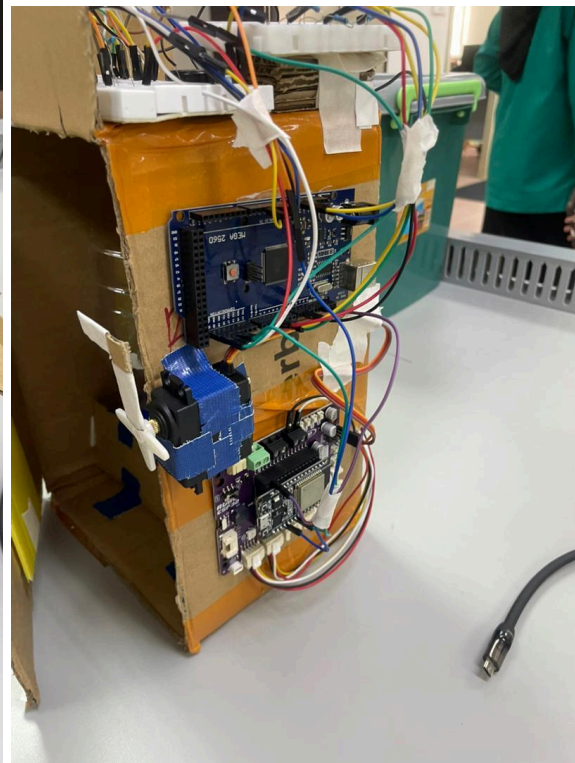
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## 1. Objective

The primary focus of this week was the physical integration of the Arduino Mega and ESP32 into the project chassis and the execution of "Stress Tests" to calibrate safety thresholds for the MPU6050 and Ultrasonic sensors.

## 2. System Assembly

- Chassis Construction: The hardware was housed in a custom enclosure with a front-facing user interface panel.
- Controller Mounting: The Arduino Mega was positioned behind the LCD to manage UI wiring, while the ESP32 was mounted near the DC motor driver to handle high-current paths.
- Sensor Calibration: The HC-SR04 Ultrasonic sensor was positioned on the door frame to detect proximity. The MPU6050 was secured to the motor base to capture real-time vibration data.





### 3. Data Testing and Calibration

Initial tests were conducted to establish "Safe" vs. "Unsafe" operating parameters. The following data was recorded:

| Test component    | Parameter tested | Recorded value        | System action                   |
|-------------------|------------------|-----------------------|---------------------------------|
| Ultrasonic sensor | Door distance    | > 15 cm               | Halt motor/ display "door open" |
| MPU6050           | Vibration force  | > 15.0 G              | Emergency stop/display "danger" |
| potentiometer     | Mode range       | 0-1023                | Map to 4 modes (A,B,C,D)        |
| DC motor          | PWM sensitivity  | 120 (min) - 255 (max) | Variable speed rotation         |

## 4. Communication Sync (Mega -> ESP32)

We verified the Serial Communication between the two boards.

- Test: Sending start/stop signals from the Mega.
- Observation: The ESP32 received commands with a latency of approximately 85 ms
- IoT Feedback: The ESP32 successfully pushed "Running" and "Error" states to the Blynk Mobile App without crashing the main motor loop.



## 5. Challenges and Solutions

- Issue: Mechanical vibration from the motor caused the Ultrasonic sensor to fluctuate wildly.
- Solution: Implemented a software "moving average" filter in the code to smooth out distance readings and prevent false "Door Open" triggers.